# GroundWalks: Reading the City Through our Feet

SAMUEL GREEMAN, Northeastern University, USA
SAMANTHA TRACEY, Northeastern University, USA
KRISTIAN KLOECKL, Northeastern University, USA
ERIC FOLMAR, Northeastern University, USA

Abstract here. Finish last. Keywords: Walkable cities, walking for health, city mapping, urban planning.

CCS Concepts: • **Human-centered computing → Ubiquitous and mobile computing**.

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## 1 INTRODUCTION

Walking is one of the primary forms of human locomotion and is practical in urban environments. Cities have also grown increasingly keen on promoting walking as a healthy form of exercise and emissions free mode of transportation that generates many beneficial interactions and outcomes espoused under the walkable cities umbrella. Walking briskly for at least one hour per day has been associated with a decrease in the expression of genes related to weight gain [15] [17]. In postmenopausal women, those who walked 7.5 miles per week had higher mean bone density of the whole body, particularly of the legs and trunk regions, than women of the same demographic who only walked less than 1 mile per day [12]. And any aerobic exercise, including walking, benefits the brain. Aerobic exercise is associated with higher levels of brain derived neurotropic factors (BDNF), a protein associated with increasing cognitive function and decreasing depression and anxiety [19].

Walkability in cities has become more of a mainstream topic of urban planning in recent decades. Jane Jacob's revolutionary ideas on walkable neighborhoods, diversity, and mixed-use elements introduced walkability into the public discourse [10]. Walkability has many definitions, but broadly we can think of it as the network of city components (including essential components such as a grocery store or a pharmacy, and social elements such as a public green space) that are easily accessible by foot in a safe and comfortable manner. Michael Southworth proposes 6 criteria for what makes a city walkable: (1) Connectivity of path network, both locally and in the larger urban setting; (2) Linkage with other modes: bus, streetcar, subway, train; (3) Fine grained and varied land use patterns, especially for local serving uses; (4) Safety, both from traffic and social crime; (5) Quality of

Authors' Contact Information: Samuel Greeman, greeman.s@northeastern.edu, Northeastern University, Boston, Massachusetts, USA; Samantha Tracey, tracey.sa@northeastern.edu, Northeastern University, Boston, Massachusetts, USA; Kristian Kloeckl, k.kloeckl@northeastern.edu, Northeastern University, Boston, Massachusetts, USA; Eric Folmar, e.folmar@northeastern.edu, Northeastern University, Boston, Massachusetts, USA.

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path, including width, paving, landscaping, signing, and lighting; (6) Path context, including street design, visual interest of the built environment, transparency, spatial definition, landscape, and overall exportability [20].

Thinking about these elements that make up walkability, there is one element that does not have reliable experience data: Quality of path.

Quality of has implications for accessibility. Interacting with variable ground conditions can pose a serious challenge for those who need a mobility aid. Mobility aids such as wheelchairs, canes, walkers, crutches, etc. are notoriously difficult to use in uneven ground conditions. Because of this, there have been city-wide initiatives to sense and map ground conditions in order to make urban environments more accessible. In Boston, residents can use the free 311 service to report non-emergency information such as potholes and sidewalk damage [1] [4]. Urban planning guidelines for the City of Boston also encourage the city to make their walking paths out of materials that are "smooth, stable, and slip resistant, with minimal gaps, rough surfaces, and vibration-causing features" [3, p. 21]. These initiatives have done what they intended to do—they have created flat and stable commuting paths that make moving throughout the city more accessible.

However, walking on a variety of ground textures has a significant impact on our health. For those who are are able-bodied, walking on uneven ground produces changes in gait and uses different leg muscles compared to even ground. In a treadmill study, those who walked on the uneven treadmill had an increase in variability of the leg muscles, particularly the thigh muscles, a greater metabolic expenditure, and had shorter, more variable gaits than those who walked on the even treadmill [21]. Similar results are found in real world walking conditions. Walking on soft, uneven ground such as grass or wood chips produces a greater energy expenditure and metabolic rate than a hard, even surface such as a sidewalk [11]. For older adults, uneven ground can create a greater fall risk; however, walking on uneven ground also helps with improving stability and coordination, leading to healthier aging as a result. When compared to younger adults, walking on uneven surfaces created changes in knee-hip and ankle-knee coordination in healthy older adults, indicating a more cautious gait strategy and attempt to maintain walking posture that can potentially reduce the risk of falls via tripping or losing balance [9].

As a way to accommodate accessibility, while also allowing for walking on variable ground textures, cities should incorporate paths that allow for more individual choice for the walker. For example, the Katy Trail in Dallas, Texas, which has a 12-foot-wide cement path, and also an 8-foot-wide soft-texture path that is pedestrian only (See Figure 1) [7]. Paths like this allow for more individual walker choice; allowing for a walker to seek out variable ground if they choose.

However, there is a lack of quantifiable data about individual ground exposure that can inform the design and implementation of such paths. Current sources for data on urban walking (manual surveys, satellite imagery and its derivatives, cameras, smartphone data, GIS-based network analysis) have limits in granularity, coverage, privacy, or ability to capture the ground-level experience of walking. GPS-based smartphone tracking of pedestrians captures location and route information but is prone to GPS-accuracy that can range from 5 to 20 meters and spike to 50 in dense urban environments [CITE]. It captures pedestrian location but is limited in assessing pedestrian ground interactions. Camera- and infrastructure-based sensing in public spaces faces privacy concerns, fixed installation and elevated cost of installation, operation, and maintenance. Satellite image analysis of ground conditions is limited by obstructions (tree canopy, built structures) and the lack of data on actual pedestrian exposure to environmental conditions. Prior insole and foot-sensing work has mostly been clinical with a focus on gait analysis for physiotherapy and recovery.

In this paper we present a new method to better understand the ground conditions urban walkers are exposed to by employing a new type of insole pressure and IMU sensor. Shoe sensors have been experimented with for the last two decades [CITE MIT PROJECT] but have been severely limited by the clunky size of technology components that interfered with regular walking and the granularity of data collected. In this project we employ a new sensor type that packs a fabric-based pressure layer, IMU unit, battery with wireless charging coil, and Bluetooth module into the regular form factor of a shoe insole resulting in no obstruction or distraction during

Fig. 1. The Katy Trail in Dallas, Texas. On the left is the hard concrete path; on the right is the red, soft-surface path. A walkway of this kind lets users choose the ground texture they seek out. Image courtesy of Friends of the Katy Trail [7].

walking activity. These insoles have been developed by our technology partner Stappone for biomechanics analysis for physiotherapy and professional athletics coaching in clinical environments, not for urban-scale sensing. Our project has successfully experimented with the use of these insole sensors for outdoor use, to better understand the ground conditions urban walkers are exposed to, and a process to make the data useful for urban planning processes.

With the insoles we collect foot pressure data from 12 distinct pressure zones per foot as well as 3-axis acceleration and gyroscope data at 100 Hz. The battery and internal data storage allows for multi-hour recording sessions. Based on this data we developed a machine learning algorithm that is capable to detect the specific ground type a walker is walking on (concrete/asphalt, pavers, gravel/compact earth, grass). Our algorithm also determines the level of evenness/unevenness, softness/hardness, and consistency/variability of every step recorded.

We then use these ground indicators to plot walking routes that encode the above metrics and compare the obtained route data with ground cover data available in today's GIS datasets. The result of our work is that we manage to determine the ground type with up to [XXX] accuracy. Furthermore, this method of capturing ground type and condition [REST OF SENTENCE]

This is the first time this kind of insole is employed at an urban-scale to detect ground type and conditions. In this proof-of-concept we collected 21,213 steps over 218 walking minutes from 4 participants that are members of the research team. This is a first step to develop a reliable method, ground detection algorithms, data visualization and GIS integration. Deliberately out of scope of this first step is a large-scale deployment, long-term wear and large sample with diverse populations.

With this project, we aim to address the following research questions and objectives:

- Can wearable insole signals reliably detect ground type and other key metrics across walking conditions?
- How can variable ground features be mapped onto geographic maps?
- How does the individual exposure to ground conditions compare and complement existing GIS data on ground cover?

This study aims to use wearable insoles as a way of measuring the variability of the different grounds that people walk on to allow us to create a quantifiable data set as an additional layer of mapping in urban environments. To our knowledge, no other study has used ergonomic wearable insoles to determine ground type in an urban environment. This added level of mapping will allow for cities to understand the types of ground people walk on in order to accommodate all types of walkers and people that use urban paths.

## 2 RELATED WORK

### 2.1 Wearable Sensing and Past Literature

Wearable sensors also provide a means of analyzing ground conditions. Insole sensors have successfully been used to measure foot pressure distribution while walking to estimate sidewalk surface conditions using a supervised learning model on four types of surface conditions: asphalt, icy, snowy, and foot-packed snow [18]. They have also successfully been used distinguish between six different floor types with 82% accuracy: sand, lawn, paving stone, carpet, linoleum, and tartan [14].

## 3 SENSORY HARDWARE AND DATA COLLECTION

In this study, members of the lab wore insoles provided by Stappone and went for walks around various parts of Boston. They also GPS recorded their walks for comparisons between ground truth and what GPS mapping claims the walked path's texture is.

### 3.1 Materials

Walking was recorded with commercial instrumented insoles (Figure 2). We deliberately built the pipeline on an off-the-shelf sensing platform rather than a bespoke array, so that the method rests on a reproducible, commercially available device and could in principle be adopted beyond our lab.

Each insole integrates **twelve plantar pressure sensors** spanning the heel, midfoot, and forefoot (Figure 3), so the insole yields a coarse *spatial* pressure map of the sole rather than a single aggregate force; this is what lets us characterize *where* and *how* load is distributed within a step, not merely how hard the foot strikes. Each insole also houses an Inertial Measurement Unit (IMU) of **two accelerometers and one gyroscope**: over three orthogonal axes the accelerometers measure linear acceleration—capturing impact and surface-transmitted vibration—while the gyroscope measures angular velocity about the same axes, capturing rotation of the foot through the stride. A magnetometer is physically present but was left disabled for this study, appearing in the exported data only as zeroed channels. Pressure and IMU are complementary: pressure describes the foot–ground contact, the accelerometer captures surface-transmitted impact and micro-vibration, and the gyroscope captures foot orientation and rotational dynamics—signals we later show, rather than raw force, are what separate compliant from firm and even from uneven ground.

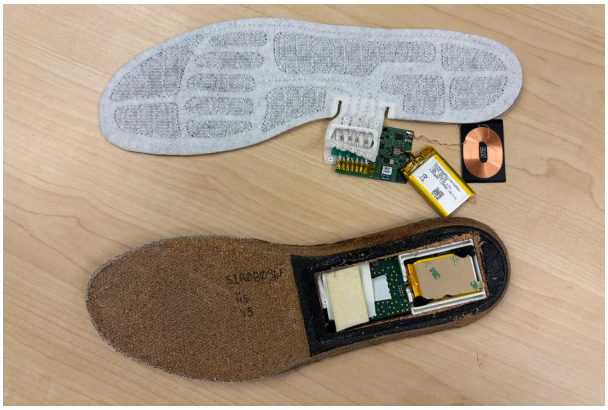


Fig. 2. Commercial instrumented insole used in this study.

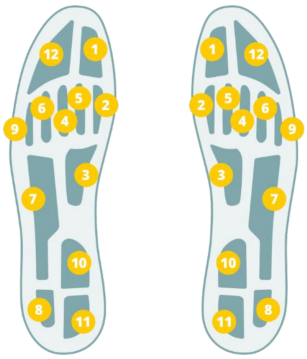



Fig. 3. Diagram of the 12 pressure sensor locations.

*Signal characteristics.* The pressure sensors report *relative* pressure quantized to 10 bits (values 0–1023); they are not calibrated to absolute force. The accelerometers span ±4000 mg at 10-bit resolution. We recorded at 62.5 Hz; the device supports higher, adjustable rates (down to a 4 ms time resolution), but 62.5 Hz was sufficient for our analysis, whose frequency-domain features occupy a sub-5 Hz band well within the corresponding Nyquist limit. Two properties of the signal directly shaped the feature design (Section 4). First, because pressure is *relative* and uncalibrated, no feature is defined on absolute pressure magnitude; every amplitude-bearing quantity is expressed as a ratio, rate, distribution shape, or session-normalized value, so that model inputs do not depend on device gain, body weight, or shoe. Second, the sampling rate bounds the vibration band we can observe; our texture features therefore target the low-frequency surface chatter that is reliably captured at 62.5 Hz rather than higher-frequency content the device could only resolve at faster rates.

Physically, the insole has a leather top layer and a cork base for comfort and durability over extended walking, and is powered by a 0.36 A h lithium-ion cell (approximately ten hours of recording). The battery and electronics, including a Bluetooth Low Energy (BLE) 4.2 module, sit beneath the heel; the radio operates in the 2.4 GHz ISM band (2.402–2.480 GHz) across 40 channels with 2 MHz spacing. Data are recorded on-board during a walk—no live Bluetooth link is required mid-recording—and exported as CSV, one row per sample per foot, comprising the

Table 1. Participants (convenience sample; project team).

| Participant | Age (yr) | Height (m) | Insole size |
|---|---|---|---|
| User 1 | 49 | 1.79 | EU 44 (US 10) |
| User 2 | 27 | 1.78 | EU 43 (US 9) |
| User 3 | 24 | 1.73 | EU 40 (US 9.5) |
| User 4 | 21 | 1.54 | EU 36 (US 5.5) |

Table 2. Per-participant gait descriptors (self-selected pace).

| Participant | Cadence (steps/min) | Velocity (m/s) | Gait cycle (s) | Stride (cm) |
|---|---|---|---|---|
| User 1 | 104.0 | 1.34 | 1.16 | 158.0 |
| User 2 | 123.9 | 1.64 | 0.97 | 161.9 |
| User 3 | 112.2 | 1.44 | 1.10 | 163.1 |
| User 4 | 117.5 | 1.27 | 1.02 | 131.0 |

twelve pressure channels, three accelerometer and three gyroscope axes, a Unix millisecond timestamp, and a sole identifier (1 = right, 2 = left).

*3.1.1 Hardware.* All processing and model training ran offline on a single desktop workstation with an 8-core AMD Ryzen 7 9700X CPU, 32 GB of DDR5 RAM, an NVIDIA RTX 5080 GPU and a 1 TB NVMe SSD, running Windows 11 and Python 3.14 with NumPy 2.5, pandas 3.0, scikit-learn 1.9, XGBoost 3.4, PyTorch 2.13 and GeoPandas 1.1. The processing pipeline is described in Section 4.

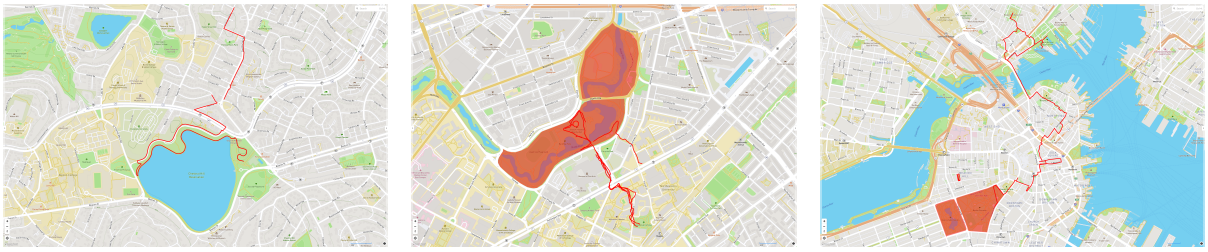
## 3.2 Participants

Walking data were collected from four participants, all members of the project team, recruited as a sample of convenience (Table 1). The group spans a range of ages (21–49 years), statures (1.54–1.79 m), and foot sizes (EU 36–44), giving some diversity in body mass, stride, and plantar loading despite the small size. All participants walked at a self-selected, comfortable pace as if walking normally rather than to a metronome or target speed.

Per-participant gait descriptors, computed from the recordings, are reported in Table 2. Walking speed varies by roughly 30% across the group (1.27–1.64 m $s^{-1}$), with correspondingly different cadences (104–124 steps/min) and stride lengths (131–163 cm); stride length tracks stature, as expected (the shortest participant has the shortest stride). This spread is methodologically relevant: walking speed is a known confound for many gait-derived features, so a model that must generalize *across* participants is also implicitly required to generalize across walking speeds. Our evaluation protocol (leave-one-user-out cross-validation, Section 5) enforces exactly this—every reported result is measured on a participant whose data, at their own walking speed, was entirely withheld during training and feature selection.

## 3.3 Data Collection Procedure

In this study, participants put the insoles into a pair of their own shoes, replacing the insole of that shoe. On their own time, participants went on walks around Boston. Walks ranged from 3-5 minutes long and included walking on a variety of ground surfaces. Before beginning a walk, the weather conditions were noted and the insoles were properly connected to the app, which recorded the walks. Participants were encouraged to make sure their phone



(a) Brighton (b) Back Bay Fens (c) Downtown (Freedom Trail)

Fig. 4. The three areas where GPS-tracked walks were recorded, all by one participant. Red lines are the walked routes in which exact GPS traces were recorded; shaded areas are regions in which walks without exact GPS or GPX data. Map data © OpenStreetMap contributors; basemap © MapBox.

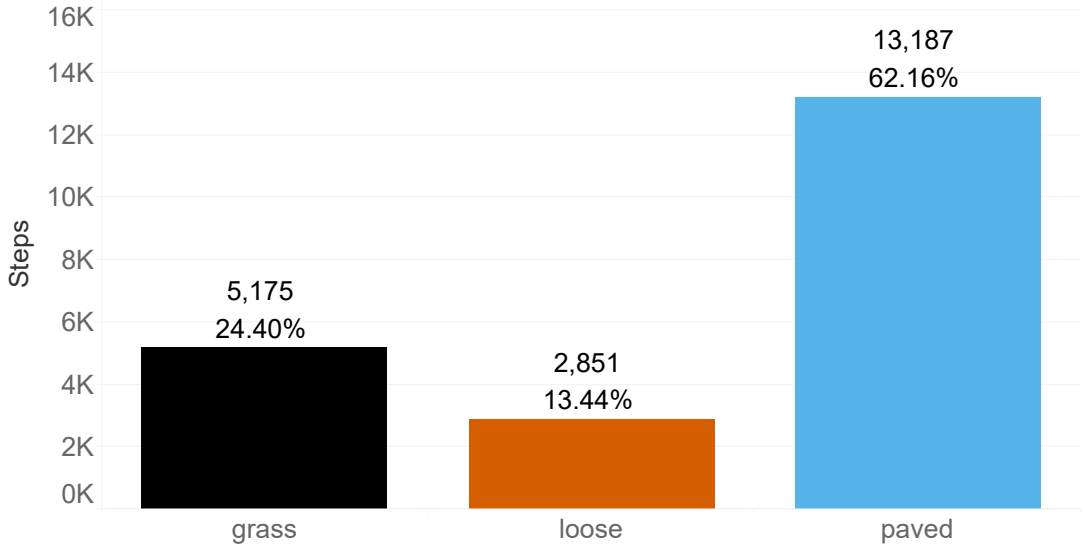



Fig. 5. Steps per surface after the step gate: 13,187 paved (62.2%), 5,175 grass (24.4%) and 2,851 loose (13.4%), for 21,213 steps in total.

batteries were fully charged, and that no other Bluetooth devices were actively paired with their smartphones to ensure a stable connection with the insoles. Participants also GPS recorded the walks via smart watch.

During the walks, participants were instructed to not constantly switch surfaces every step and to avoid stepping on sewers, drains, manhole covers, and other manmade surfaces (such as the yellow textured pads on curb cuts). After each walk, participants noted which shoes they were wearing and uploaded their recordings to the app. They also shared their GPS data from their walk with the research team.

Walks with GPS tracks were recorded in three areas of Boston (Figure 4), all by one participant. After the step gate the dataset contains 21,213 steps (Figure 5), though participants walked different surface mixes (Figure 6), so between-participant comparisons can partly reflect where each person walked.

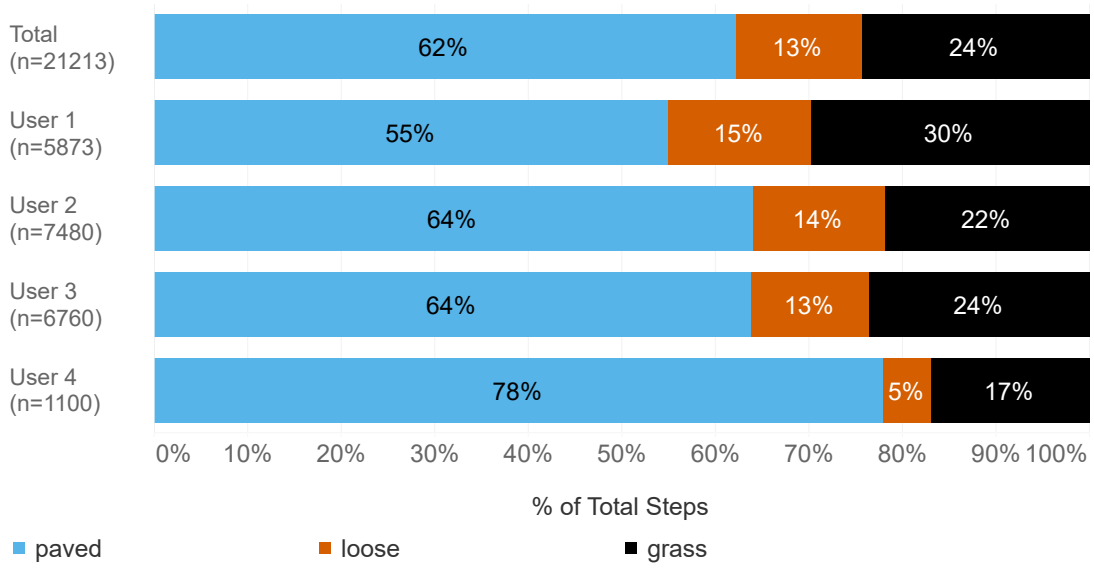



Fig. 6. Share of steps on each surface, by participant and overall. Participants walked different surface mixes, so between-participant differences can partly reflect where each person walked.

## 4 DATA PIPELINE ENGINEERING

Each recording is a continuous multichannel walk sampled at $f_s = 62.5$ Hz: twelve relative plantar pressure channels $p_1(t), \ldots, p_{12}(t)$, tri-axial acceleration $\mathbf{a}(t) = (a_x, a_y, a_z)$, and tri-axial angular velocity $\boldsymbol{\omega}(t) = (\omega_x, \omega_y, \omega_z)$ per foot (Section 3.1). The heel pads $(p_8, p_{11})$ and forefoot pads $(p_2, p_4, p_5, p_6, p_9)$ are grouped anatomically; all twelve contribute to whole-foot measures. Each recording is accompanied by a ground-contact event file exported by the insole software, which supplies the step boundaries (Section 4.2). Surface extents and non-surface events are marked inline by the walker (Section 4.3).

Figure 7 summarizes the processing chain described in this section.

### 4.1 Recording-level quality gate

Before any feature is computed, each raw recording passes an automated quality gate. A recording is rejected if it has fewer than 100 rows, does not contain both soles, or has more than 5% missing values across the IMU and pressure channels. The gate additionally reports (without rejecting) flat-lined channels, out-of-range pressures ($< 0$ or $> 5000$), corrupt-flagged rows, and large timestamp gaps ($> 3\times$ the median inter-sample interval). Only passing recordings are compiled into the dataset.

### 4.2 Step segmentation from ground-contact events

Steps are delimited by foot-contact events exported by the insole software alongside each recording. For each sole, the ground-contact file lists the Unix-millisecond time of every initial contact (heel strike), $t_k^{\mathrm{HS}}$, and of the subsequent toe-off, $t_k^{\mathrm{TO}}$. Each *step window* runs from one initial contact to the next of the same foot,

$$W_k = \left[t_k^{\mathrm{HS}},\ t_{k+1}^{\mathrm{HS}}\right), \tag{1}$$

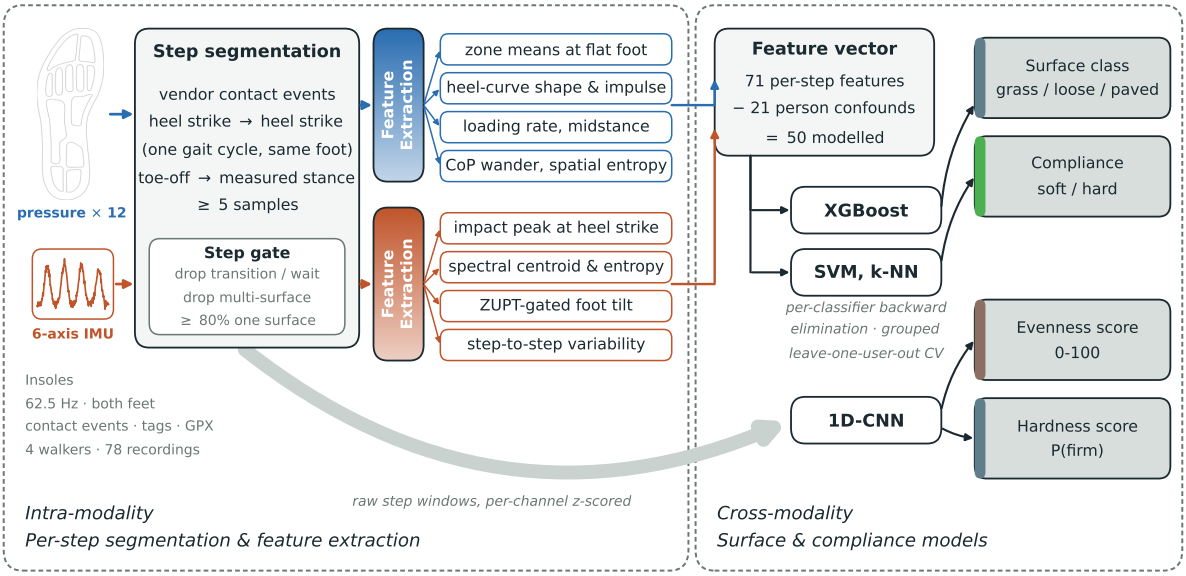



Fig. 7. Processing pipeline. Each step window runs from one heel strike to the next heel strike of the same foot, taken from the vendor's ground-contact events, and passes the step gate. Pressure and IMU features are computed per step; excluding 21 person-dependent confounds leaves 50 features, from which backward elimination selects a subset for each classifier. The 1D-CNN bypasses feature extraction and learns from the raw step windows. Classifiers are compared under grouped leave-one-user-out cross-validation.

mapped to sample indices by the first sample at or after each event time. A window therefore spans one complete gait cycle of that foot—a *stride* in gait-analysis terms; we use "step" throughout for brevity. Because the window starts at contact, the stance and swing phases fall in the same order and the same place within every window. The paired toe-off additionally gives a *measured* stance time $T_k^{\mathrm{st}} = t_k^{\mathrm{TO}} - t_k^{\mathrm{HS}}$ and stance fraction $T_k^{\mathrm{st}}/(t_{k+1}^{\mathrm{HS}} - t_k^{\mathrm{HS}})$; the window length itself is the gait-cycle duration, retained under its historical feature name `stance_duration` (Table 3).

Windows shorter than 5 samples are discarded, and events that resolve to the same sample are merged. A sole with fewer than two events inside its recorded span contributes no windows, and a recording without an event file is excluded rather than segmented by another method, so that no dataset mixes two window definitions. All per-step features are computed within a single window.

*Why contact events rather than a pressure-peak detector.* An earlier version of the pipeline delimited steps at local maxima of the smoothed heel-pressure sum, $\tilde{h} = \mathrm{SG}_{w,q}[p_8 + p_{11}]$ with $w = 11$, $q = 2$, taking peaks at least 50 samples apart with prominence $\geq 200$ and height $\geq 500$. Compared against the contact events, those peaks lag initial contact by a median of 112 ms to 128 ms (s.d. 13 ms to 16 ms) across both feet: the heel-sum maximum marks the loading-response peak, not contact. A peak-to-peak window consequently began roughly 12% into the gait cycle, placing each step's loading ramp at the *end* of the preceding window. Features defined relative to the window start were the most affected. The time-to-heel-peak fraction measured $0.43 \pm 0.49$ under peak anchoring—effectively undefined—against $0.125 \pm 0.025$ under event anchoring, consistent with the expected timing of the loading-response peak; and the heel rise and fall slopes, which under peak anchoring were computed from a window that began at the peak itself, went from a coefficient of variation above 2 to 0.1–0.3. Features

anchored to the within-window heel minimum, such as loading rate and impact, were essentially unchanged. Event anchoring also removes three tuned detector constants from the method. On a check set of three recordings (≈ 1,300 steps) the two segmentations differ in step count by under 1%, so dataset composition is effectively unchanged. The peak detector remains available in the released pipeline as a switchable alternative.

The contact events are produced by the vendor's software from the same pressure signal, using processing that is not public. They are therefore a better-anchored segmentation rather than an independent measurement of foot contact, and their behavior is tied to the version of the export software.

### 4.3 Surface labeling and the step gate

Inline markers assign each row a surface by carrying a marker forward until the next marker. Two marker types are non-surface events—`transition` and `wait`—which open an *excluded* region rather than a surface. A step window is admitted only if it is cleanly on one surface: it is dropped if it (i) touches any excluded row, (ii) spans two different surface labels (straddles a change), or (iii) is less than 80% covered by a single surface label (a partial or segment-edge step). A surviving window takes the majority surface of its rows. This gate is applied identically to the feature pipeline and to the raw-window sequence model; because both also cut their windows from the same contact events, the two operate on identical step windows. At prediction time (no labels) the gate is bypassed.

### 4.4 Session normalization

To remove between-session offsets (body weight, shoe, insole seating), amplitude-bearing quantities are referenced to session-level anchors computed once per recording. With total pressure $s(t) = \sum_i p_i(t)$ over loaded samples $\mathcal{L} = \{t : s(t) > 0\}$, we form $\mu_s, \sigma_s$ (mean/std of $s$ on $\mathcal{L}$) and $\mu_a, \sigma_a$ (mean/std of $|a_y|$). Impact is standardized as $(|a_y| - \mu_a)/\sigma_a$; loading rate and impulse are divided by $\mu_s$. All other features are ratios or distribution shapes and are inherently scale-free.

### 4.5 Foot orientation via a zero-velocity (ZUPT) gate

Surface unevenness is read from foot orientation at the flat-foot instant of each step. Within a step we form the gyroscope magnitude $g(t) = \sqrt{\omega_x^2 + \omega_y^2 + \omega_z^2}$ and take the flat/stationary samples to be the lowest quartile of angular rate, $\mathcal{S} = \{t : g(t) \le Q_{25}(g)\}$. There the accelerometer measures predominantly gravity, so foot tilt follows from the gravity direction as two inclination angles

$$\theta_A = \arctan2\big(a_x, \sqrt{a_y^2 + a_z^2}\big), \qquad \theta_B = \arctan2\big(a_z, \sqrt{a_x^2 + a_y^2}\big). \tag{2}$$

Per step we keep the in-contact tilt range $\Delta\theta = \max_{\mathcal{S}} \theta - \min_{\mathcal{S}} \theta$ (foot rock). The *absolute* resting tilt encodes only the sensor-mount offset and is discarded; what is retained is its step-to-step variability (below), a direct proxy for surface unevenness.

### 4.6 Spectral and roughness descriptors

Within-step vibration is characterized from a signal $x(t)$ (e.g. $a_y$) after linear detrending ($x' = x - (\alpha t + \beta)$) so the stance loading curve does not dominate. With power spectrum $P(f) = |\mathrm{FFT}[x']|^2$ and $P_{\mathrm{tot}} = \sum_f P(f)$, we compute the (amplitude-invariant) spectral centroid, high-frequency fraction ($f_c = 5\,\mathrm{Hz}$), and normalized entropy:

$$\text{centroid} = \frac{\sum_f f P(f)}{P_{\mathrm{tot}}}, \quad \text{hf_frac} = \frac{\sum_{f \ge f_c} P(f)}{P_{\mathrm{tot}}}, \quad H = \frac{-\sum_f \hat{p}(f) \log \hat{p}(f)}{\log N_f}, \;\; \hat{p} = \tfrac{P}{P_{\mathrm{tot}}}. \tag{3}$$

Table 3. Per-step features by group (names as computed in the pipeline).

| Group | Features |
|---|---|
| Step timing (4) | `stance_duration` (gait-cycle duration; see Section 4.2), `stance_time_sec`, `stance_frac_true`, `time_to_peak_frac` |
| Impact & loading (5) | `impact_mag`, `impact_mag_norm`, `accel_mag_strike`, `loading_rate`, `loading_rate_norm` |
| Heel/forefoot pressure (8) | `heel_pmax_ratio`, `fore_pmax_ratio`, `heel_mean_ratio`, `fore_mean_ratio`, `heel_fore_ratio`, `total_pmax`, `total_p_mean`, `midstance_diff` |
| Heel-curve shape (7) | `heel_impulse`, `heel_impulse_norm`, `rise_slope`, `fall_slope`, `slope_asym`, `heel_skew`, `heel_kurt` |
| Gyroscope (7) | `gyro_mag_strike`, `gyro_{x,y,z}_strike`, `gyro_{x,y,z}_std` |
| Total-pressure shape (8) | `gu_dip_depth`, `gu_load_first_frac`, `gu_push_peak`, `gu_load_slope`, `gu_unload_slope`, `gu_fore_early_frac`, `gu_fore_heel_toff`, `gu_stance_frac` |
| Spatial distribution (6) | `gu_spatial_entropy_ff`, `gu_spatial_entropy_stance`, `gu_participation_ff`, `gu_active_sensors`, `gu_cop_wander`, `gu_press_jitter` |
| Foot orientation (4) | `orient_range_a`, `orient_range_b`, `orient_rest_a_roll3_std`, `orient_rest_b_roll3_std` |
| Texture & frequency (7) | `accel_spec_centroid`, `accel_hf_frac`, `accel_spec_entropy`, `accel_zcr`, `accel_jerk_ratio`, `gyro_hf_frac`, `heel_ripple_frac` |
| Step-to-step (3) | `impact_mag_roll3_std`, `loading_rate_roll3_std`, `stance_duration_roll3_std` |
| Per-sensor means (12) | `pressure_01_mean_ratio` through `pressure_12_mean_ratio` |

Two further scale-free roughness proxies are the zero-crossing rate of $x'$, the jerk-to-signal ratio, and a ripple fraction:

$$\frac{\mathrm{RMS}(\Delta x)}{\mathrm{RMS}(x')}, \qquad \frac{\sum_t \left(x - \mathrm{SG}_{w,q}[x]\right)^2}{\sum_t x^2},$$

where $\Delta x(t) = x(t) - x(t-1)$. The ripple fraction isolates residual micro-vibration after the smooth loading trend is removed (e.g. mortar-joint chatter on rigid textured ground). The 62.5 Hz rate places the analyzed band ($< 5$ Hz) well within the Nyquist limit.

## 4.7 Step-to-step variability

For a per-step scalar $u^{(k)}$ we compute a rolling three-step standard deviation, $\mathrm{roll3}(u)^{(k)} = \mathrm{std}\{u^{(k-2)}, u^{(k-1)}, u^{(k)}\}$, applied to resting tilt $(\theta_A, \theta_B)$, impact, loading rate, and stance duration. A high value means the quantity jumps from stride to stride—the hallmark of walking on irregular ground.

## 4.8 Per-step feature set

From each admitted step window we compute a fixed vector spanning eleven groups (Table 3). Timing and impact features capture *when* and *how hard* the foot lands, including the measured stance time and stance fraction taken from the contact events; pressure-distribution and heel-curve-shape features capture *where* load sits and the shape of the loading profile; gyroscope and total-pressure-curve features capture foot rotation and the stance profile; twelve-pad spatial features treat the sole as a pressure map (distribution entropy, effective number of loaded pads, center-of-pressure wander, per-pad jitter). Two groups target the axes that best separate surfaces: the texture/frequency and ZUPT-fusion descriptors defined above (spectral shape, roughness, in-contact tilt range, and step-to-step tilt variability). In total 71 features are computed per step.

### 4.9 Person-confound removal

Of the 71 computed features, 21 are excluded from every model because cross-user permutation analysis showed they act as a person fingerprint rather than a surface cue: the twelve per-pad normalized means and nine raw-magnitude / heel–forefoot-mean features (total pressure, raw impact, heel/forefoot mean and peak ratios, raw impulse). These inflate in-sample importance but degrade generalization to a held-out walker, so they are removed after computation, leaving 50 features for modeling. Any feature with zero variance across the dataset, or that duplicates another column exactly, is additionally removed before modeling. Every remaining amplitude feature is a normalized ratio or a variability/shape measure, so no model input scales with body weight or shoe. Per-classifier feature *selection* is described in the model section. All parameters above are fixed constants in the released pipeline, so feature extraction is deterministic given a recording.

## 5 MACHINE LEARNING FOR GROUND DISCRIMINATION

Our goal in this section is to establish what a step, measured at the foot, reveals about the ground it landed on. We pose this as two supervised classification tasks over the same step windows, and add two continuous per-step indices that describe ground character within a class.

The *three-class* task assigns each step to the surface underfoot: *grass* (lawn and other vegetated ground), *loose* (gravel, stone dust, compacted dirt and other unbound mineral surfaces), or *paved* (concrete, asphalt, brick and other bound surfaces). These three divide the walkable ground of an urban park or street network, and they are the classes a GIS ground-cover layer attempts to record (Section 6.2). The *two-class* task collapses them by compliance, *soft* (grass) against *hard* (loose and paved), asking the easier question of whether ground yields underfoot. We report the three-class task first and the two-class task second, since the latter is a coarsening of the former.

Beyond class, two attributes vary *within* a class and matter for how a surface is to walk on: how firm it is (Section 5.4) and how evenly a person can step on it (Section 5.5). A gravel path and a stone-dust path are both loose, but need not be equally even. These two indices, together with the surface class, are what Section 6 places on a map.

### 5.1 Evaluation Regimes

We use two grouped cross-validation schemes, which answer different questions. *Leave-one-user-out* (LOUO) withholds all of one participant's recordings, so every prediction concerns a person the model has never seen. *Leave-one-recording-out* (LORO) withholds one recording at a time, so each walk is predicted by a model that never saw that walk but has seen the walker's other recordings; since each mapped walk is one recording, LORO is equivalently leave-one-walk-out. Under neither scheme does a model see the data it is scored on.

LOUO is the stricter test and the one we use to compare classifiers and select features. With four participants, however, it is an average over four held-out people and cannot be read as performance on a new person from the wider population; we report it as the hardest available test on this cohort, not as evidence of general transfer. LORO matches the situation in the applications that follow: all GPS-tracked walks, and therefore every walk in the GIS comparison (Section 6), were recorded by one participant whose other recordings are in the dataset, so those maps characterize that participant's walks. We report both side by side, and read the gap between them as how much performance depends on having seen the walker before.

Feature subsets were selected by backward elimination on LOUO macro-F1 rather than on in-sample fit, which favors features that transfer between people over features that fit one participant. The selection is not independent of either evaluation: it was scored on the LOUO folds and used every recording, including each walk later held out under LORO. Selected-feature scores under both regimes are therefore an upper bound, and the full-feature LOUO scores the conservative estimate.

Table 4. Three-class grouped leave-one-user-out results. "Full-50" uses all non-confound features; "Selected" uses each classifier's subset. The CNN uses raw windows (no feature selection). The majority row always predicts *paved*, the most frequent class. Macro-F1 is the primary metric.

| | Full-50 | | | | | Selected | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Macro | F1 | | | | Macro | F1 | | | |
| Model | Acc. | F1 | grass | loose | paved | Acc. | F1 | grass | loose | paved | #feat |
| XGBoost | **0.737** | 0.528 | **0.698** | 0.055 | **0.831** | **0.723** | 0.575 | **0.707** | 0.196 | **0.822** | 22 |
| SVM-linear | 0.588 | 0.502 | 0.572 | 0.218 | 0.715 | 0.655 | **0.583** | 0.690 | **0.297** | 0.763 | 33 |
| SVM-poly | 0.639 | 0.482 | 0.571 | 0.107 | 0.767 | 0.634 | 0.534 | 0.639 | 0.203 | 0.759 | 33 |
| k-NN | 0.664 | 0.401 | 0.396 | 0.018 | 0.788 | 0.712 | 0.524 | 0.668 | 0.093 | 0.812 | 9 |
| CNN | 0.671 | **0.549** | 0.619 | **0.258** | 0.771 | – | – | – | – | – | raw |
| Majority | 0.622 | 0.256 | 0 | 0 | 0.767 | | | | | | |

## 5.2 Three-Class (Grass, Loose, Paved) Ground Discrimination

The three-class task separates loose ground (gravel, stone dust, dirt) from both paved and grass surfaces. All scores are from grouped leave-one-user-out cross-validation (*source_user*), our primary metric, in which each fold is tested on a participant withheld from training. We compare five classifiers: four feature-based models (XGBoost, linear and polynomial SVM, *k*-NN), each on the full 50-feature set and on its own subset selected by backward elimination (Section 5.2.2), and a 1D-CNN operating on raw step windows, which has no feature-selection variant.

*5.2.1 Classifier comparison.* Table 4 reports the results. With selected features, the linear SVM (macro-F1 0.583) and XGBoost (0.575) lead, followed by the polynomial SVM (0.534) and k-NN (0.524). The two leaders differ by less than one macro-F1 point, well within the variability of a four-participant cross-validation, so we treat them as comparable; they trade off differently, XGBoost reaching the highest accuracy (0.723) and the linear SVM the best loose-class F1 (0.297). Every model clears the majority baseline's macro-F1 of 0.256 by a wide margin, but accuracy alone is a poor guide here: the polynomial SVM's selected-feature accuracy (0.634) is only 1.2 points above that of always predicting *paved* (0.622). Without feature selection, the CNN learning from raw windows is the strongest three-class model (macro-F1 0.549, loose F1 0.258), feature selection lifts the linear SVM and XGBoost above it. The CNN does not repeat this on the two-class task (Section 5.3), where it trails every feature-based model.

Loose ground is the hard class for every model. Its F1 is at most 0.297, against 0.64–0.71 for grass and 0.76–0.82 for paved among the selected-feature models. Loose is the rarest class, and plausibly the least distinct mechanically: a compacted stone-dust path is firm like pavement but rougher, while loose gravel yields underfoot more like grass. The gap between this task and the binary one (Section 5.3) is concentrated in this one class: grass F1 (0.64–0.71) is close to the two-class soft-class F1 (0.64–0.73).

*5.2.2 Effect of feature selection.* Feature selection raised macro-F1 for every feature-based model (Figure 8), again most for the distance-based k-NN (0.401 → 0.524, +0.123) and the linear SVM (0.502 → 0.583, +0.081), and least for the polynomial SVM (+0.052) and XGBoost (+0.047). The gains are concentrated in the minority classes.

k-NN's grass F1 rose from 0.396 to 0.668, and loose F1 rose for every model—for XGBoost from 0.055 to 0.196, which is the difference between ignoring loose ground and detecting some of it. Selection slightly lowered XGBoost's accuracy (0.737 → 0.723) and paved F1 (0.831 → 0.822): with all features it leaned harder on the majority class. Selection also narrows the spread between models (full-feature macro-F1 0.40–0.53, selected 0.52–0.58).

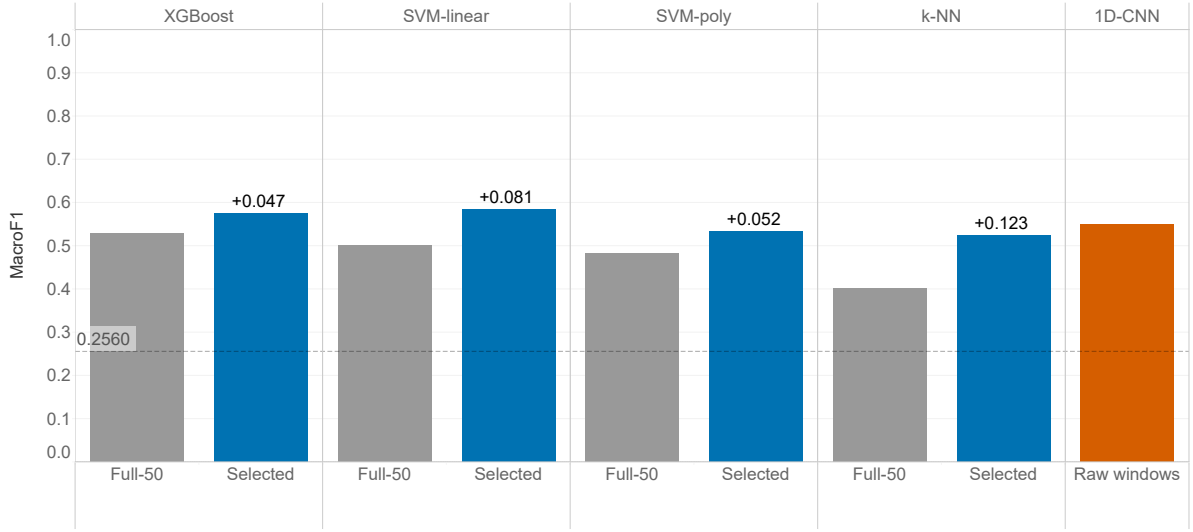



Fig. 8. Three-class macro-F1 with all 50 features and with each classifier's selected subset, under grouped leave-one-user-out cross-validation. The CNN learns from raw windows and has no selected variant. The dashed line marks the majority-class baseline (0.256). Selected scores are an upper bound, because the elimination was guided by the same cross-validation (Section 5.3.2).

Each classifier's subset was selected by backward elimination: starting from all 50 features, each step removed the feature whose removal gave the highest cross-user macro-F1, stopping once the feature to be removed had a positive cross-user permutation importance. The elimination was guided by the same leave-one-user-out macro-F1 that we report, so the selected scores are optimistic: over its steps the procedure compares many candidate subsets on the same four held-out participants and keeps the best-scoring path, and part of each gain reflects fitting those particular folds. We therefore treat the full-feature scores as the conservative estimate of cross-user performance and the selected scores as an upper bound.

*5.2.3 A shared feature core.* Six features were retained by all four feature-based classifiers: two measures of step-to-step foot-tilt variability (`orient_rest_a_roll3_std`, `orient_rest_b_roll3_std`), the accelerometer jerk ratio (`accel_jerk_ratio`), high-frequency energy fraction (`accel_hf_frac`) and spectral entropy (`accel_spec_entropy`), and roll-rate variability (`gyro_x_std`). Nine more were retained by three of the four, among them the measured stance time from the contact events (`stance_time_sec`), center-of-pressure wander (`gu_cop_wander`) and stance-duration variability. Four features were retained by none (`gu_active_sensors`, `gu_spatial_entropy_ff`, `gyro_y_std`, `heel_fore_ratio`). Every core feature is an inertial measure of vibration or steadiness; none is a pressure feature. Distinguishing three surfaces, like distinguishing two, rests on how the foot moves and shakes on contact rather than on how hard it presses. Figure 9 shows the most separable of these features by class.

*5.2.4 Interpretation.* Three-way surface discrimination is learnable across participants, but less reliably than the binary task: the best models reach macro-F1 around 0.58, against 0.822 for soft versus hard, and the loss is almost entirely in recognizing loose ground. The discriminative signal is a compact set of inertial steadiness and vibration features. These figures describe performance on this four-participant cohort rather than an estimate of population-level generalization (Section 7.5).

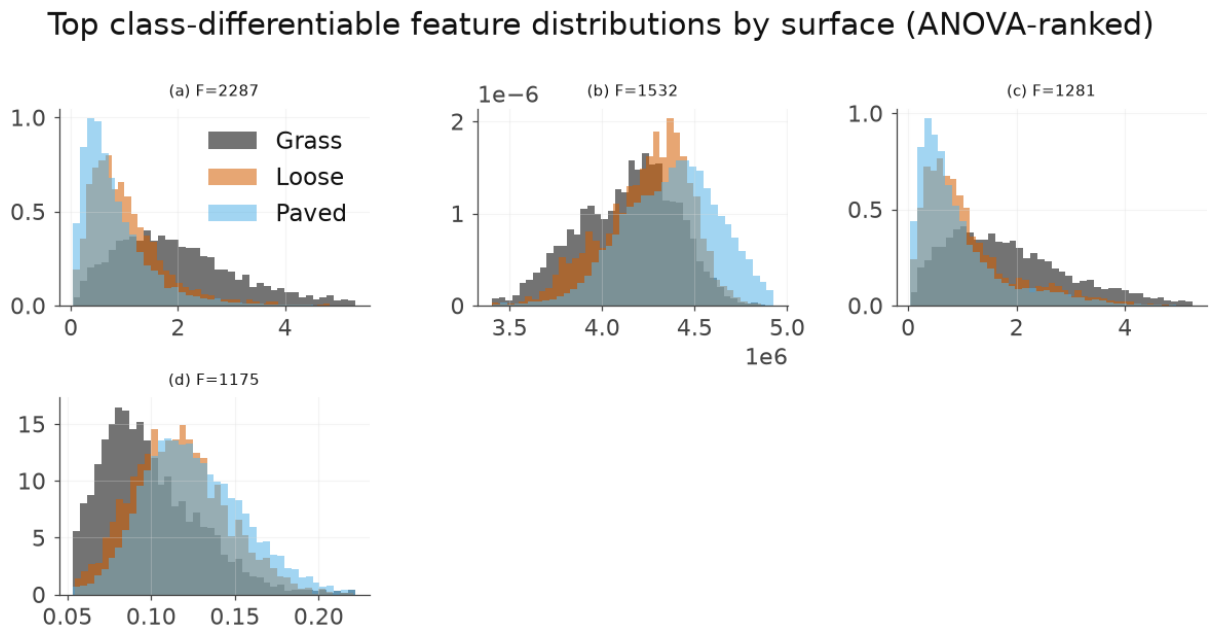



Fig. 9. Distributions of the four features that best separate grass, loose and paved ground, over all steps: `orient_rest_a_roll3_std`, `gyro_x_std`, `orient_rest_b_roll3_std` and `accel_hf_frac`, in panel order. All four belong to the three-class core. Panels are ordered by one-way ANOVA $F$, which here only orders the panels (see Figure 11).

*5.2.5 Known-walker performance (LORO).* Table 5 repeats the evaluation under leave-one-recording-out, the regime that matches the mapped walks. Every model keeps its LOUO-selected feature subset; we did not re-select under LORO, so these are not optimized for this regime and no full-feature comparison is reported. Loose-class F1 roughly doubles for the best model, from 0.297 under LOUO to 0.554, and the ranking changes: the CNN becomes the strongest model by a wide margin (macro-F1 0.781 against XGBoost's 0.647), having been middling under LOUO.

That reversal is the most notable feature of the table. The CNN gains +0.232 macro-F1 from LORO, while the four feature-based models gain between −0.028 and +0.072. A model that is unremarkable across walkers but dominant within a walker is behaving as though much of what it learned is walker-specific: the raw-window representation appears to capture an individual's gait signature alongside the surface signal, where the engineered features, being summary statistics of a step, capture less of it. This makes the CNN the better choice when a target walker's data is available for training and the worse choice when it is not, and it is consistent with the CNN's weak LOUO showing. We note it as an observation about this cohort rather than a general property of the architecture.

## 5.3 Two-Class (Soft vs. Hard) Ground Discrimination

Collapsing the three surfaces by compliance gives the binary task: compliant (grass) against firm (paved and loose) ground. The protocol, classifiers and feature-selection procedure are those of Section 5.2; only the labels change.

Table 5. Three-class grouped leave-one-recording-out (LORO) results, using each classifier's LOUO-selected subset and the CNN's raw windows. Each walk is predicted by a model trained on every other recording, including the walker's other walks. LOUO macro-F1 from Table 4 is repeated for comparison.

| | | Macro | F1 | | | Macro-F1 | |
|---|---|---|---|---|---|---|---|
| Model | Acc. | F1 | grass | loose | paved | LOUO | Δ |
| CNN | **0.854** | **0.781** | **0.887** | **0.554** | **0.903** | 0.549 | +0.232 |
| XGBoost | 0.771 | 0.647 | 0.793 | 0.294 | 0.856 | 0.575 | +0.072 |
| k-NN | 0.740 | 0.569 | 0.731 | 0.143 | 0.835 | 0.524 | +0.045 |
| SVM-linear | 0.562 | 0.555 | 0.732 | 0.299 | 0.633 | 0.583 | −0.028 |
| SVM-poly | 0.557 | 0.555 | 0.759 | 0.298 | 0.609 | 0.534 | +0.021 |

Table 6. Two-class grouped leave-one-user-out results. "Full-50" uses all non-confound features; "Selected" uses each classifier's subset. The CNN uses raw windows (no feature selection). The majority row always predicts *hard*, the more frequent class. Macro-F1 is the primary metric.

| | Full-50 | | | | Selected | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | Macro | F1 | | | Macro | F1 | | |
| Model | Acc. | F1 | soft | hard | Acc. | F1 | soft | hard | #feat |
| XGBoost | **0.826** | **0.783** | **0.686** | **0.880** | **0.868** | **0.822** | **0.732** | **0.913** | 12 |
| *k*-NN | 0.788 | 0.610 | 0.347 | 0.873 | 0.842 | 0.781 | 0.667 | 0.896 | 10 |
| SVM-poly | 0.784 | 0.722 | 0.590 | 0.854 | 0.823 | 0.760 | 0.637 | 0.883 | 30 |
| SVM-linear | 0.640 | 0.627 | 0.556 | 0.698 | 0.762 | 0.734 | 0.646 | 0.821 | 31 |
| CNN | 0.753 | 0.715 | 0.612 | 0.818 | – | – | – | – | raw |
| Majority | 0.756 | 0.431 | 0 | 0.861 | | | | | |

*5.3.1 Classifier comparison.* Table 6 reports the results. With selected features XGBoost is the strongest model on every metric: macro-F1 0.822, accuracy 0.868, soft-class F1 0.732 and hard-class F1 0.913. It is followed by *k*-NN (0.781), the polynomial SVM (0.760) and the linear SVM (0.734). XGBoost's lead over the next model (0.041 macro-F1) is larger than the gaps among the others, but with four participants the ordering below it should not be over-read. The soft class is harder than the hard class for every model (selected-feature F1 0.64–0.73 against 0.82–0.91), reflecting its minority share of 24.4% of steps.

Accuracy must be read against the majority baseline of 0.756, the accuracy of always predicting *hard*. The linear SVM's selected-feature accuracy (0.762) barely exceeds it, and its full-feature accuracy (0.640) falls below it; its macro-F1 is nonetheless well above the baseline's 0.431 in both cases, because it recovers the soft class at the cost of more errors on the hard one. The CNN, learning from raw windows, reaches 0.715 macro-F1, below XGBoost's 0.783 on the same full feature set: in this cohort the engineered features carry more cross-user signal than the network extracts from the raw signal.

*5.3.2 Effect of feature selection.* Each classifier's subset was selected by backward elimination. Starting from all 50 features, each step removed the feature whose removal gave the highest cross-user macro-F1, stopping once the feature to be removed had a positive cross-user permutation importance.

Selection raised macro-F1 and accuracy for every feature-based model, by an amount that tracks each model's sensitivity to input dimensionality. The gain was largest for *k*-NN (0.610 → 0.781, +0.171) and the linear SVM (0.627 → 0.734, +0.107), and smallest for XGBoost (+0.039) and the polynomial SVM (+0.038). The *k*-NN

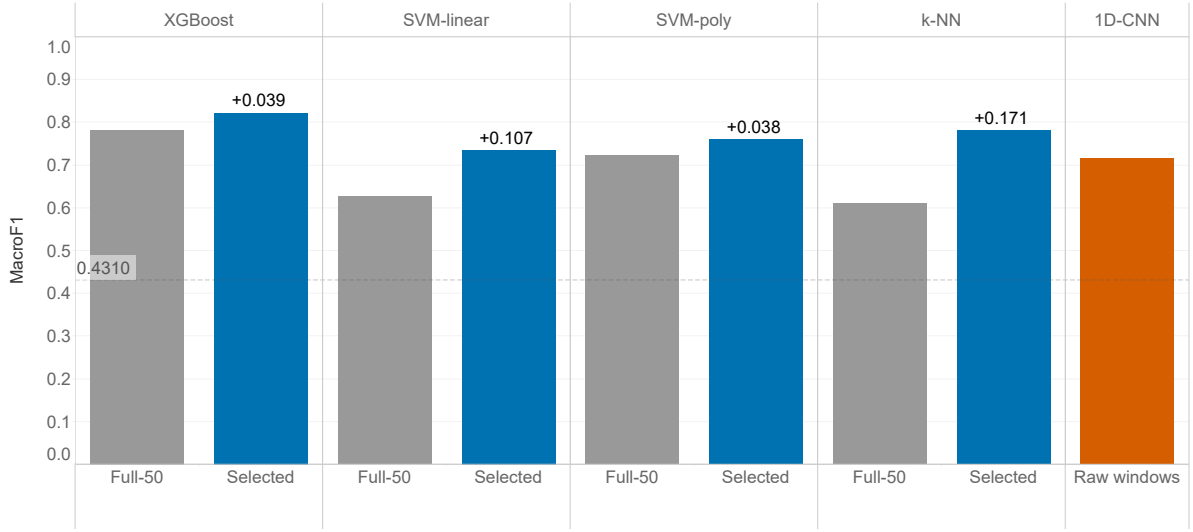



Fig. 10. Two-class macro-F1 with all 50 features and with each classifier's selected subset, under grouped leave-one-user-out cross-validation. The CNN learns from raw windows and has no selected variant. The dashed line marks the majority-class baseline (0.431). Conventions as in Figure 8.

improvement is almost entirely recovery of the soft class, whose F1 rose from 0.347 to 0.667: with all 50 features the distance metric was likely dominated by dimensions irrelevant to surface, and it largely missed the minority class, and pruning to ten features restored it. Selection also narrows the spread between models, from 0.61–0.78 macro-F1 with all features to 0.73–0.82 with selected ones (Figure 10).

These selected-feature scores carry a caveat. The elimination was guided by the same leave-one-user-out macro-F1 that we report, so the selected scores are optimistic: over its steps the procedure compares many candidate subsets on the same four held-out participants and keeps the best-scoring path, and part of each gain reflects fitting those particular folds. We therefore treat the full-feature scores as the conservative estimate of cross-user performance and the selected scores as an upper bound.

*5.3.3 A shared feature core.* Eight features were retained by all four feature-based classifiers: three measures of foot tilt—its step-to-step variability about both axes (`orient_rest_a_roll3_std`, `orient_rest_b_roll3_std`) and its range within the step (`orient_range_b`)—the accelerometer jerk ratio (`accel_jerk_ratio`), high-frequency energy fraction (`accel_hf_frac`) and spectral entropy (`accel_spec_entropy`), roll-rate variability (`gyro_x_std`), and gait-cycle duration (`stance_duration`). Two more were retained by three of the four (`accel_spec_centroid`, `accel_mag_strike`), and nine by none: two spatial pressure-distribution measures (`gu_participation_ff`, `gu_spatial_entropy_ff`), four measures of the heel's loading curve (`loading_rate`, `time_to_peak_frac`, `slope_asym`, `heel_kurt`) and three gyroscope measures (`gyro_y_std`, `gyro_y_strike`, `gyro_hf_frac`). Every core feature is an inertial measure of vibration or steadiness, or gait timing; none is a pressure feature. Soft and hard ground are distinguished by how regularly and smoothly a person walks on them, not by how hard they press. XGBoost reaches its scores with twelve features and $k$-NN with ten. This core contains the three-class core entire (Section 5.2.3), adding only tilt range (`orient_range_b`) and gait-cycle duration (`stance_duration`): the same six inertial features carry both tasks. Figure 11 shows how the most separable of these features are distributed for each class.

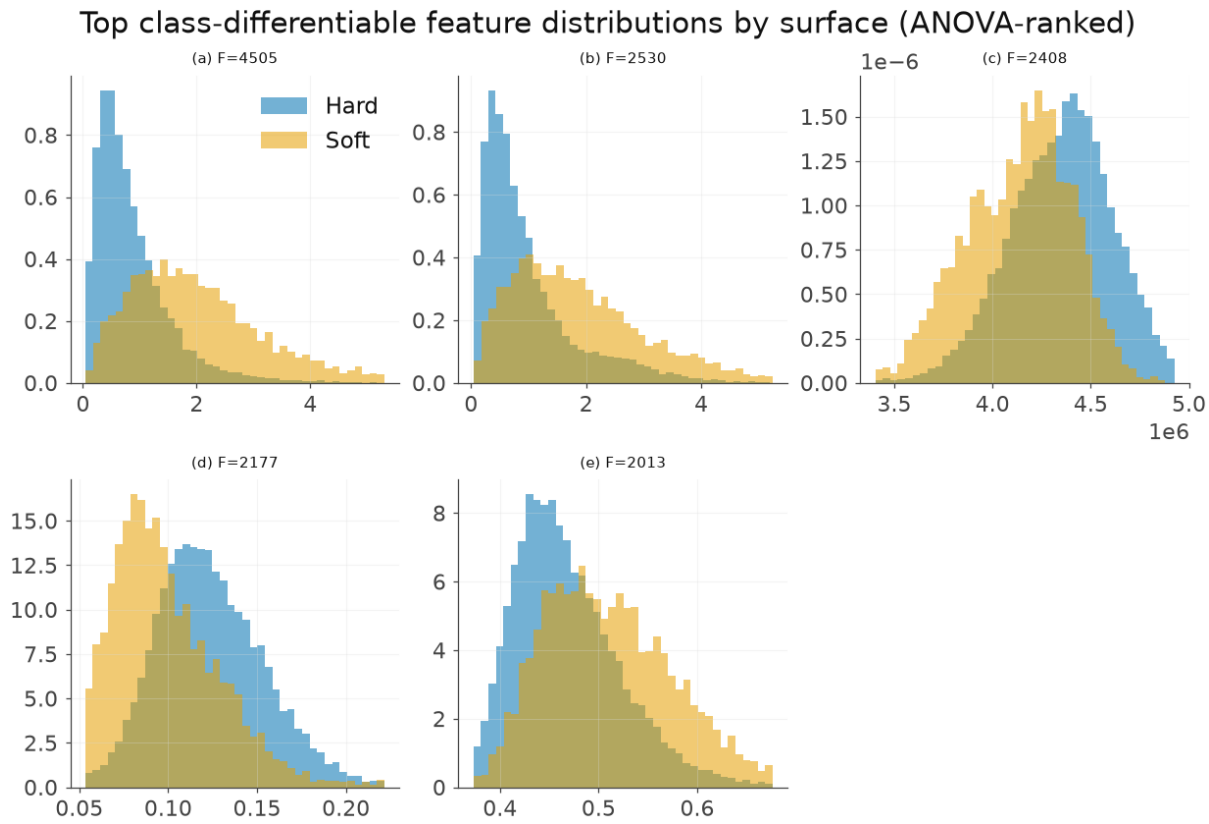



Fig. 11. Distributions of the five features that best separate soft from hard ground, over all steps: `orient_rest_a_roll3_std`, `orient_rest_b_roll3_std`, `gyro_x_std`, `accel_hf_frac` and `accel_jerk_ratio`, in panel order. All five belong to the shared feature core, which contains the three-class core entire (Section 5.2.3), adding only tilt range and gait-cycle duration. Panels are ordered by one-way ANOVA $F$, shown in each panel; because steps from one participant are not independent, $F$ only orders the panels and is not a significance test.

*5.3.4 Interpretation.* The binary task is learnable across participants: the best model reaches macro-F1 0.822 under leave-one-user-out evaluation, selection benefits every feature-based model and especially the distance-based one, and the discriminative signal is low-dimensional and physically interpretable. The residual soft–hard gap reflects the compliant class's minority share. All results are on a fixed four-participant cohort; we therefore report them as performance *on this cohort* rather than as an estimate of population-level generalization, and treat the small sample as the principal limitation (Section 7.5).

*5.3.5 Known-walker performance (LORO).* Table 7 gives the same evaluation under leave-one-recording-out, again with each classifier's LOUO-selected subset and no re-selection. The ordering mirrors the three-class task: the CNN leads at macro-F1 0.928 against XGBoost's 0.862, after trailing every feature-based model under LOUO, and its gain of +0.235 is again several times that of any other model (+0.037 to +0.073). For a known walker, soft-versus-hard discrimination reaches 0.95 accuracy; for an unseen walker it does not exceed 0.87.

Table 7. Two-class grouped leave-one-recording-out (LORO) results, using each classifier's LOUO-selected subset and the CNN's raw windows. LOUO macro-F1 from Table 6 is repeated for comparison.

| | | Macro | F1 | | Macro-F1 | |
|---|---|---|---|---|---|---|
| Model | Acc. | F1 | soft | hard | LOUO | Δ |
| CNN | **0.946** | **0.928** | **0.892** | **0.964** | 0.693 | +0.235 |
| XGBoost | 0.896 | 0.862 | 0.793 | 0.931 | 0.822 | +0.040 |
| k-NN | 0.873 | 0.818 | 0.718 | 0.918 | 0.781 | +0.037 |
| SVM-linear | 0.837 | 0.807 | 0.732 | 0.883 | 0.734 | +0.073 |
| SVM-poly | 0.833 | 0.805 | 0.732 | 0.879 | 0.760 | +0.045 |

## 5.4 Hardness: Probability of Firm Ground

Surface class is categorical, but compliance is not: a well-compacted stone-dust path is firmer underfoot than loose gravel, though both are *loose*. We therefore also derive a continuous *hardness* index as the probability of firm ground, from a logistic regression on eight compliance-related features computed from the same step window (Section 4.2): accelerometer spectral centroid, high-frequency energy fraction, jerk ratio and zero-crossing rate, gyroscope high-frequency fraction, stance fraction, gait-cycle duration, and normalized loading rate. The probability is scaled to 0–100 for display.

Hardness is trained on the same soft/hard labels as the two-class task, so it is a graded reading of that task rather than independent of it; its value lies in resolving *how* firm a step was within a class, not in adding information the class does not already carry. Because it is label-trained, any per-surface summary of hardness must come from held-out predictions, or it reports separation the model was fitted to produce.

## 5.5 Evenness: An Unsupervised Index of Stepping Steadiness

The two classification tasks and the hardness index all require surface labels. The *evenness* index does not: it is computed without reference to any label, and is included because the attribute that matters most to a walker—whether the ground can be stepped on steadily—is not what a surface is made of.

*Construction.* For each step we compute ten features in two families. Five describe stability *within* the step: the range of foot tilt about each of two axes, center-of-pressure wander across the twelve sensors, the jitter of the pressure signal, and gyroscope variability. Five describe variability *between* consecutive steps, as the rolling standard deviation over three steps of stance duration, loading rate, impact magnitude, and tilt at rest about each axis. Each feature is standardized robustly, using the median and interquartile range of the fitting set rather than mean and standard deviation, so that a few irregular steps do not set the scale. The ten standardized values are clipped to ±4, averaged into a single deviation score, and mapped through a normal cumulative distribution function to 0–100, with higher values denoting steadier stepping. The transformation is fitted once over the whole dataset and stored, so that scores from different walks are on one scale.

*What it does and does not measure.* Because no labels enter the computation, the index can be applied to any walk, including surfaces absent from our training data, and it cannot inherit the classifier's biases. The same property limits its interpretation: the index measures the steadiness of the *walking*, which is a joint product of the ground and the walker. A person walking cautiously on smooth pavement and a person walking confidently on gravel may receive similar scores. Figure 12 shows the distribution by ground-truth surface; the overlap between classes is substantial, which is consistent with the index capturing something other than surface identity. We therefore present evenness as a descriptive per-step metric rather than a validated measure of surface condition, and treat its validation against an independent measure of surface roughness as future work (Section 7.5). We report no classification scores for it, since it predicts no label.

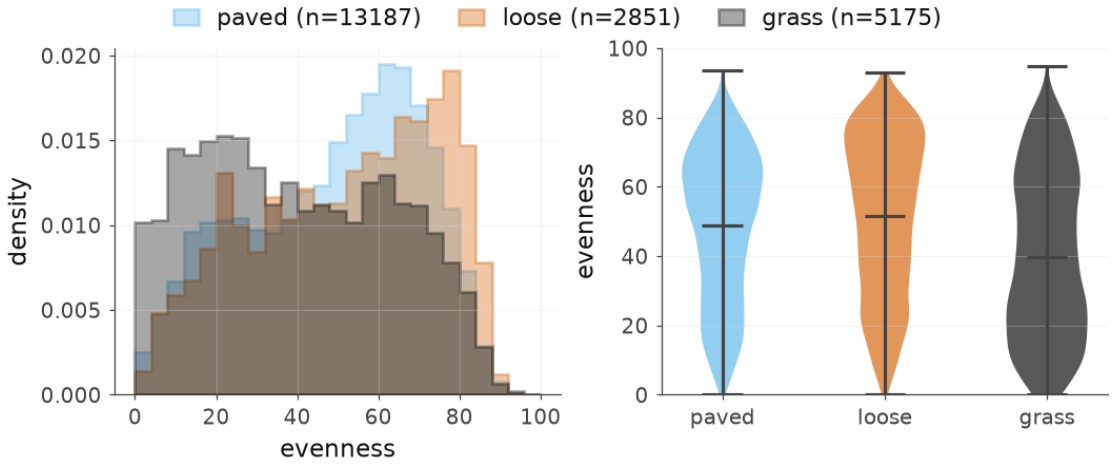



Fig. 12. Distribution of the per-step evenness index by ground-truth surface, over all steps: (left) histogram, (right) violin plot with horizontal bars at the median. The index is unsupervised: no surface labels enter its computation.

## 6 MAPPING GROUND CONDITIONS: COMPLEMENTING GIS GROUND-COVER DATA

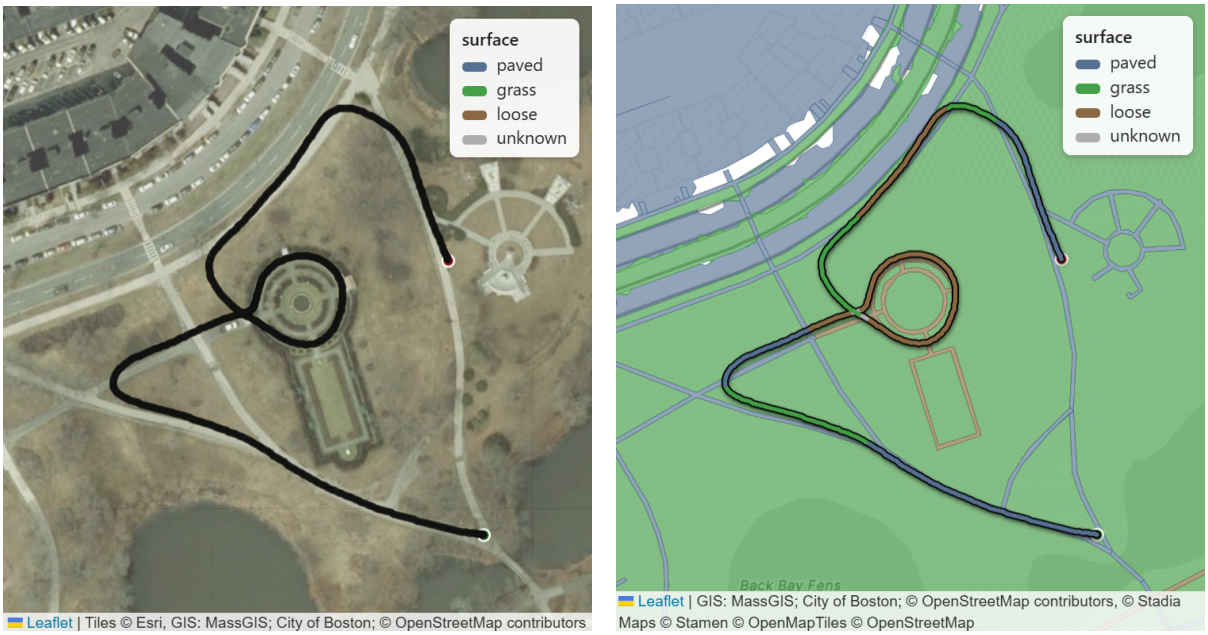
The models of Section 5 label individual steps. Placed on a map, those labels become a record of the ground *as it was walked*, a different kind of information from the ground-cover layers that cities and open-data platforms already publish. This section describes how several per-step ground attributes are mapped together (Section 6.1), how the result compares with four public GIS sources for the same walks (Section 6.2), and how the two could be combined in a GIS-based planning workflow (Section 7.4).

### 6.1 From steps to maps

*Per-step attributes.* Each admitted step carries three attributes from Section 5, all computed from the same step window: its *surface class* (grass, loose, paved), its *hardness* index (Section 5.4) and its *evenness* index (Section 5.5). The first is categorical; the latter two are continuous on 0–100 and are rendered with diverging color ramps.

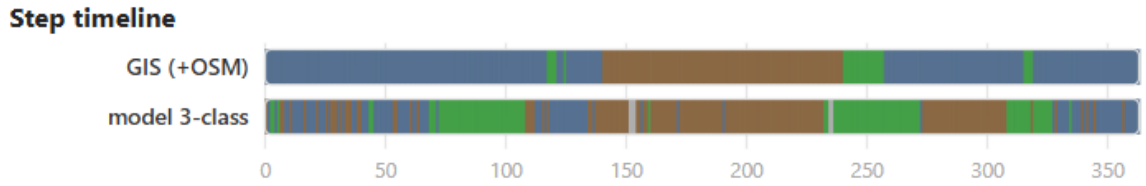
*Georeferencing.* A step is timestamped at its initial contact and assigned to the nearest fix of the concurrently logged GPS track, provided one lies within 1.5 s; otherwise it is left unplaced rather than interpolated. Consecutive fixes define line segments, each inheriting the attributes of the step assigned to its start. Spatial resolution along the route is therefore one value per stride—1.31 m to 1.63 m across our participants (Table 2)—while positional accuracy is bounded by consumer GPS, typically several meters. The maps resolve *which stretch* of a route had a given character, not position within the width of a sidewalk.

*Showing several attributes at once.* Figure 14 shows the evenness and hardness layers for the same walk. Mapping one attribute per map hides the relationships that matter, such as whether a stretch the model calls paved is also uneven. We instead draw each attribute as its own line, offset perpendicular to a smoothed copy of the track so the lines run in parallel along a single shared geometry (Figure 13). Any subset of the ground truth, the GIS-derived label, the model's class, hardness, and evenness can be shown together, and every line is co-registered step for step. A linked timeline below the map shows the same attributes against step index; hovering a step on the timeline locates it on the map. The ground-truth line is drawn above the others with an outline and drop shadow so it remains legible where its color matches the surface beneath.




(a) Aerial imagery

(b) Ground truth over the GIS surface map




(c) Step timeline

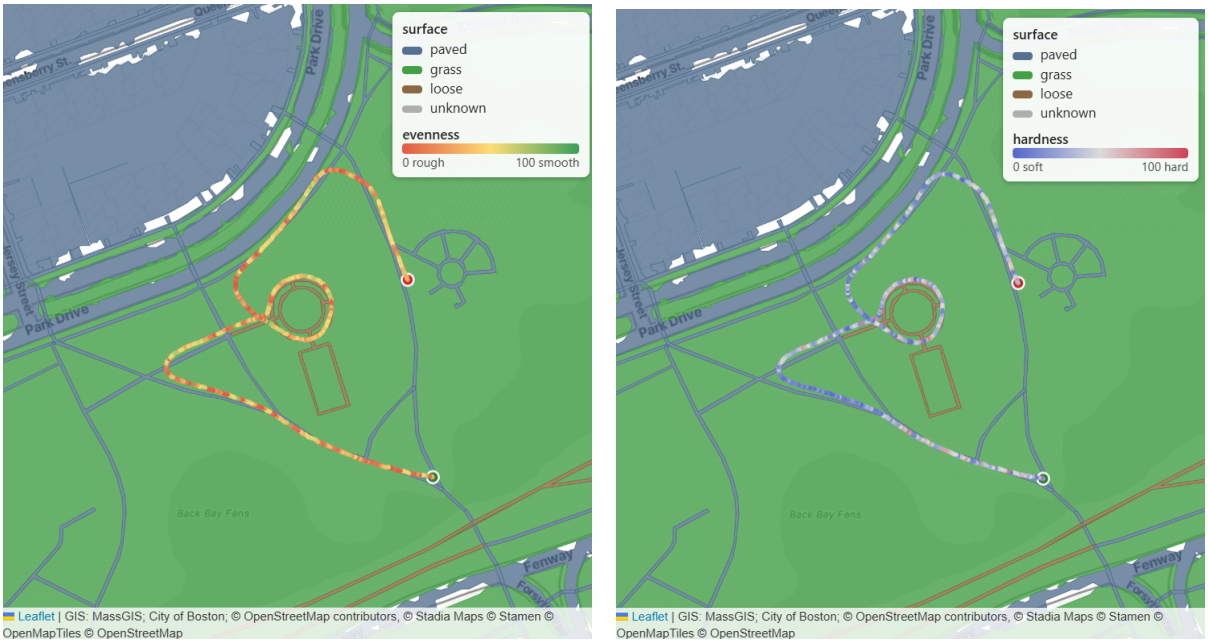
Fig. 13. Walk 155900 through the Back Bay Fens, as walked and as mapped. (a) The route over aerial imagery. (b) The walker's ground truth over the GIS surface classification: the stretches walked on loose paths lie over areas the GIS layers classify as grass or paved. (c) The ground truth, GIS label and model prediction for every step in order; model values are leave-one-recording-out predictions. Map data © OpenStreetMap contributors (ODbL); land cover: MassGIS; sidewalks and open space: City of Boston; imagery © Esri; basemap © Stadia Maps, Stamen Design, OpenMapTiles.

*Basemaps.* The same lines can be drawn over aerial imagery, which lets a reader check a label against what is visibly on the ground, or over the GIS surface classification itself, rendered as a map in its own right (Section 6.2), which makes agreement and disagreement between measured and published ground cover directly visible.

## 6.2 Comparison with public GIS ground-cover data

*Sources.* We assembled four public datasets that, between them, describe the surface of the walked routes (Table 8). They differ in what they record, how they were produced, and when.

*A GIS label per step.* To compare like with like, we derive a GIS label for every georeferenced step. For each step, every source is queried on its own and the label is taken from the highest-ranked source that returns a usable class under a configurable precedence order. A source answers only if the step's GPS fix lies within a common




(a) Evenness (b) Hardness

Fig. 14. The same walk, scored step by step: (a) evenness and (b) hardness, one value per step, on the GIS surface map of Figure 13b. Credits as in Figure 13.

Table 8. Public GIS sources used as the comparison baseline. "Underfoot" indicates whether the source describes the walking surface itself rather than land cover seen from above or a parcel's designation.

| Source | What it records | Unit | Vintage | Underfoot |
|---|---|---|---|---|
| MassGIS Land Cover/ Land Use [13] | 19 cover classes from overhead imagery, plus parcel land use | 1 m polygons, wall-to-wall | 2016 imagery (pub. 2019) | No |
| Sidewalk Inventory [6] | Sidewalk material and a visually assessed condition index | Sidewalk polygons | 2014 survey | Yes |
| Open Space [2] | Extent and type of parks and open spaces | Site polygons | Maintained | No |
| OpenStreet-Map [16] | Volunteer-tagged surface of paths and ways | Lines | Continuously edited | Yes |

reach of one of its features—3 m, comparable to the GPS error, for sidewalk and open-space polygons, plus half a path width (1 m) for OSM paths, whose distance is measured to the centerline. Land cover covers the whole area, so its reach (1.5 m) matters only at polygon edges. A shared reach keeps any one source from winning steps simply by reaching further; a larger reach labels a fix several meters into a lawn with the material of the

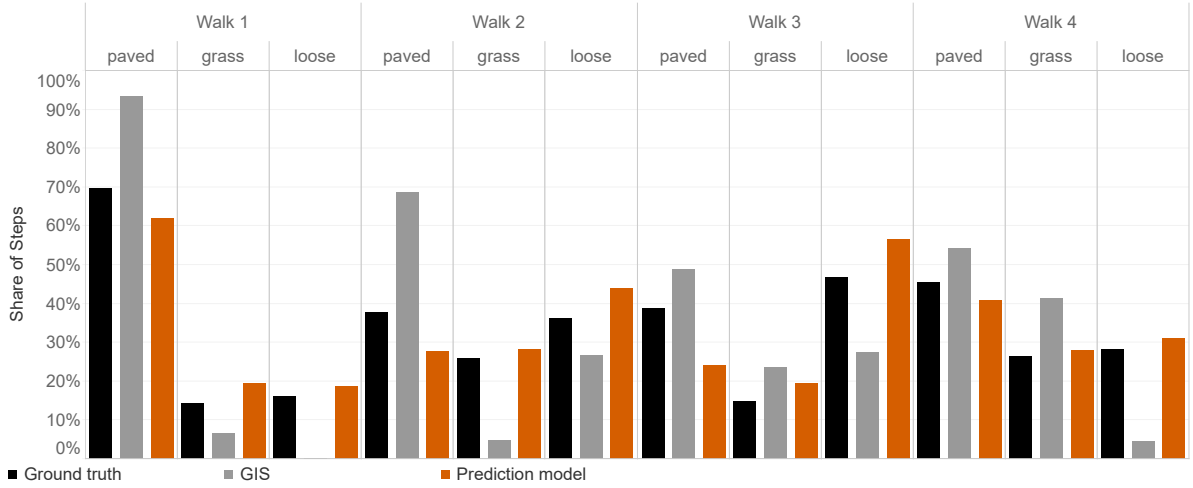



Fig. 15. Share of each walk's steps assigned to each surface by the GIS label, the walker's ground truth, and the three-class model, for the four mapped walks. Composition shows each source's bias but not per-step accuracy, which Table 9 reports. Model shares are from leave-one-recording-out predictions.

Table 9. Per-step agreement with ground truth on the mapped walks. Model predictions are leave-one-recording-out (the walk itself held out from training) and, for comparison, leave-one-user-out (all of the walker's recordings held out); the GIS label is fixed by the public data. *n* counts steps with a ground-truth surface tag.

| | | | Per-class F1 | | | Unlabeled |
|---|---|---|---|---|---|---|
| Source | Accuracy | Macro-F1 | grass | loose | paved | steps (%) |
| GIS (4 sources) | 0.733 | 0.667 | 0.504 | 0.662 | 0.834 | 1.8 |
| 3-class model | 0.802 | 0.791 | 0.779 | 0.765 | 0.829 | 0 |

nearest sidewalk. We report how GIS agreement depends on the reach in Table 8, sweeping it from 0 m to 15 m. Land-cover classes that do not correspond to a walking surface (tree canopy, shrub, water, wetland) and untagged OSM ways are treated as uninformative rather than guessed. The same precedence resolves overlapping polygons when the sources are drawn as a single surface map. Figures show each source's polygons at their published footprint; because a fix within the reach of a feature takes its label, the per-step label can differ from the map within that reach of a polygon's edge.

*Agreement with ground truth.* Table 9 compares, step for step, the GIS-derived label and the three-class model against the walker's ground-truth tags on the same steps. Model predictions are leave-one-recording-out: each walk is labeled by a model trained on every other recording, including the walker's other walks but never the walk itself (Section 5.1). All mapped walks were recorded by one participant, so the comparison characterizes that participant's walks. The GIS label is fixed by the public data; the model's feature subsets were selected on the full dataset, which includes these walks.

The surface composition of individual walks already shows where each source errs (Figure 15). On walk 155900, ground truth tagged 36.1% of steps loose, 37.8% paved and 26.1% grass. The GIS label assigns only 26.6% to loose, and over-assigns paved (68.6%); the model leans the opposite way, assigning 44.0% to loose and 27.7% to paved.

The same pattern holds on all four mapped walks: GIS under-assigns loose ground and over-assigns paved on every one, while the model's errors run consistently, if more modestly, in the opposite direction. Composition alone cannot establish accuracy—two sources can match the truth's proportions while disagreeing on every step, which is why Table 9 compares steps individually—but it locates the GIS layers' principal blind spot: loose ground on paths. Overhead land cover registers bare ground only where it is open to the sky, and a compacted-gravel or stone-dust path beneath park trees is classified by its canopy. OpenStreetMap is the only source that tags such paths directly, and coverage is thin: of 11,617 ways in our study-area extract, 29% carry no surface tag at all, and the loose-tagged ways total 9.8 km.

*What the insoles add.* Set against these sources, the insole-derived maps contribute three things.

- *Attributes the published layers do not carry.* The GIS sources describe what a surface *is*—a material, a cover class, a designation. Hardness and evenness describe how it *behaves* underfoot. The closest published analogue is the Sidewalk Inventory's condition index, a visual assessment of sidewalk defects; it covers sidewalks only, not park paths, and describes the surface rather than a walker's response to it.
- *Currency.* The sidewalk survey dates from 2014 and the land-cover imagery from 2016. An insole measurement reflects the ground on the day it was walked, and repeated walks would track change—resurfacing, construction, seasonal wear—that a static layer cannot.
- *Density along the path.* Where a published layer assigns one value to a whole sidewalk polygon or path, the maps carry one value per stride, so variation within a single block or path is retained.

The comparison runs the other way as well. GIS layers are wall-to-wall and need no one to walk them; insole data exist only where someone has. The insole signal is also, in part, a property of the walker: the person-confound analysis of Section 4 shows how strongly individual gait shapes raw measurements. The two are complementary rather than competing: GIS supplies coverage and a prior, and walked data supply experienced condition and currency where people actually go.

# 7 DISCUSSION

## 7.1 What this study contributes

Instrumented insoles are a clinical instrument. They are used in gait laboratories, in rehabilitation, and in studies of balance and fall risk, almost always indoors, over short distances, and on a surface chosen to be uniform. This study takes that instrument outdoors and applies it at the scale of a city: participants walked ordinary urban routes—park paths, sidewalks, a historic trail—and the ground beneath each step was recovered from the signals at the foot.

Around that shift we built a complete processing pipeline: step segmentation anchored to ground-contact events, a quality gate at recording and step level, 71 per-step features of which 21 are removed as person-dependent confounds, and a set of classifiers evaluated under participant-grouped cross-validation. The pipeline recovers not only the surface class but two further characteristics of the ground as walked—how firm it is and how steadily it can be stepped on—and each output is georeferenced, so that a walk becomes a map layer rather than a score.

Under leave-one-user-out evaluation, soft-versus-hard discrimination reaches macro-F1 0.822 and three-way discrimination 0.583, both far above their majority baselines, with loose ground the limiting class. For a known walker, the same models reach 0.928 and 0.781. The discriminative signal is compact and interpretable: a handful of inertial measures of vibration and step-to-step steadiness, with no pressure feature in the shared core of either task.

## 7.2 Why measure ground this way

The motivation is that cities have begun to build deliberately varied ground and have no instrument that measures what they built. Soft-surface paths beside paved ones (Section 1), permeable paving, stabilized stone dust and the

material rules in Boston's own sidewalk standards all presume that surface texture matters to pedestrians; the data layers available to the planners who specify them record material and extent, not what the ground is like to walk on. The GIS comparison in Section 6.2 shows how far apart those two things can be on the same route. Per-step measurement supplies exactly the missing quantity, at the resolution at which walking happens, and in a form existing GIS can consume.

### 7.3 On the prospect of wider sensor availability

The obvious objection is that instrumented insoles are not ubiquitous: today they are clinical and research instruments, not consumer products, and a method that depends on them cannot yet be applied at the scale urban analysis would require. We make no claim that such data exists at scale now.

The history of comparable wearable sensors suggests the constraint may be temporary. GPS-equipped sports watches, the Nike+ in-shoe accelerometer, and the inertial sensors now embedded in ski boots and running shoes each moved from specialist instrumentation to broad consumer distribution within roughly a decade, and each generated, as a side effect, large volumes of spatially referenced movement data that was subsequently used for purposes its designers had not intended—including, in the case of fitness-tracker traces, by transport planners. Pressure-sensing insoles are on a similar trajectory, driven by sports performance and diabetic foot care rather than by urban analysis.

Our purpose is therefore to establish the method before the data arrives: to show what can be recovered from instrumented-insole signals, what it adds to existing ground-cover data, and what would have to be true—in cohort size, in validation, and in privacy practice (Section 7.4)—before such measurements could responsibly inform planning decisions. [Add citations for the Nike+/GPS-watch/ski-boot adoption, and for planners' use of fitness-tracker traces]

### 7.4 Toward integration in GIS-based urban planning

Because every output is a georeferenced feature with typed attributes, it can be delivered in the formats planning GIS already consume (GeoJSON, OGC services) and loaded into the same desktop and web tools as the layers in Table 8. We see four uses.

*From traces to segment attributes.* Planning works at the level of street and path segments, not individual traces. Steps would be map-matched to a reference network—OSM way identifiers or the Sidewalk Inventory's own segment identifiers—and summarized per segment as a median and interquartile range for hardness and evenness, a class distribution for surface type, and, essential for interpretation, the number of distinct walks and walkers behind each value. A segment measured once by one person and a segment measured fifty times by twenty people should not look equally certain on a map.

*Auditing and updating published layers.* Persistent disagreement between a segment's measured class and its GIS label, such as a path the land cover calls grass that walkers consistently experience as loose, flags a candidate correction. For OpenStreetMap, such flags would be offered to human mappers for review rather than applied automatically, in keeping with the project's norms for automated edits.

*Maintenance prioritization.* Evenness hotspots on sidewalks can be compared with the inventory's condition index and with resident reports through Boston's 311 service [1], in the same spirit as Street Bump's crowd-sourced road-roughness mapping [5], but for the pedestrian network and its unpaved paths. Agreement between independent signals strengthens a case for repair; disagreement is itself informative about what each one misses.

*Accessibility-aware routing.* Segment-level hardness and evenness are natural edge costs for pedestrian routing. A walker using a cane, walker or wheelchair could be routed away from loose or uneven stretches that today's route planners present as indistinguishable from smooth pavement [8].

*Requirements before deployment.* Three conditions follow from this study's limits. The models must be validated on a larger and more varied cohort, including people who use mobility aids, before their outputs inform decisions affecting them. The evenness index, being unsupervised, needs validation against an independent measure of surface condition before it is interpreted as surface evenness rather than steadiness of gait. And walking traces are personal location data: any shared layer should be aggregated to segments, trimmed at trip origins and destinations, and released only where a minimum number of distinct walkers contribute.

## 7.5 Limitations

This study is a proof of concept, and its limits should be read as bounding what the results can be taken to show.

*Cohort.* Four participants cannot support population-level claims. A leave-one-user-out score here is an average over four held-out people, so its fold-to-fold variation is large and it cannot be read as performance on a walker drawn from the wider population. The participants are also the authors, which risks the pipeline being implicitly fit to their gait; we guard against the direct form of this by never admitting a test participant's data into training or feature selection, but not against the indirect form, in which design choices made while looking at these four people's data happen to suit them. Nor does the cohort include the people for whom ground condition matters most: no participant used a mobility aid, and age, gait characteristics and footwear vary little. The three-class results rest on a particularly thin basis for loose ground, which is the rarest class overall and is represented by only 57 steps from one participant.

*Feature selection.* Subsets were chosen by backward elimination guided by the same leave-one-user-out macro-F1 that we report, over many candidate subsets and four held-out people. The selected-feature scores are therefore an upper bound rather than an unbiased estimate; the full-feature scores are the conservative figure. Nested selection, in which features are chosen only within each training fold, would settle how much of the gain survives, and we have not run it.

*GPS accuracy in the GIS comparison.* The comparison with public GIS layers (Section 6.2) is asymmetric in a way that favors the insoles. Ground truth and model predictions come from the insole signal and are unaffected by position error, but the GIS label is looked up at the GPS fix, so a displaced fix can return the ground cover of a neighboring polygon. Across our walks, fixes lay a median of 2.9 m from the nearest mapped path centerline (per-walk medians 1.7 m to 5.4 m), and in nine of ten walks between 21% and 53% of fixes lay more than 5 m away. Since the land-cover layer is queried within 3 m, some share of the disagreement we attribute to GIS is positional error in our own track rather than error in the published data. We report the per-walk offsets alongside the agreement figures for this reason, and note that the walk we present in detail has among the cleanest GPS of the set.

*Scope of the mapped results.* All GPS-tracked walks were recorded by a single participant, so the mapping results characterize that walker's routes, not walkers in general, and are evaluated under LORO rather than LOUO for that reason.

*The evenness index.* Being unsupervised, the index is not validated against any independent measure of surface condition, and it plausibly reflects the steadiness of the walking as much as the evenness of the ground. It should be read as descriptive until validated against instrumented surface-roughness measurement.

Broadening the cohort in size and diversity, nested feature selection, and validation of the evenness index are the priorities for future work.

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Catherine, Stappone, etc.

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